MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SCH 403: LANTHANIDES AND ACTINIDES

Date: 19th September, 2018 Time 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.

SCH 403: LANTHANIDES AND ACTINIDES

Instructions

Answer Question one and any other TWO questions in section B

SECTION A [30 MARKS]

QUESTION ONE

a) Explain why the additional electron in Gd does not enter 4f- orbital but it goes to 5d level. [2 marks]

b) Explain why oxidation electrode potential values for the couple:

$Ln^0(s) \text{--->} Ln^{3+}(aq) + 3e^-$ is high positive. [1 mark]

c) Give the symbols for the following lanthanides

(i) Protactinium

(ii) Einstenium

(iii) Californium [1.5 marks]

d) Give the names of the following elements represented by the symbols:

(i) Dy

(ii) Pr

(iii) Yb [1.5 marks]

e) Complete and balance the following reactions:

(i) $Sm^{+2} + H_3O+$ [2 marks]

(ii) $Eu^{+2} + H_2O$ [2 marks]

f) What is meant by lanthanide contraction? Explain the origin of lanthanide contraction [3 marks]

g) List **five** consequences of lanthanide contraction. [5 marks]

h) Explain why Ce^{4+} is strongly oxidizing while Sm^{2+} is strongly reducing? [4 marks]

i) Explain the following observations:

(I) The colours of lanthanide ions are independent of the nature of the surrounding ligands. [3 marks]

(II) Lanthanide ions only form organometallic compounds with good donor ligands. [3 marks]

j) Give two uses of elemental lanthanides [2 marks]

SECTION B [40 marks]

Answer any TWO questions from this section

QUESTION TWO [20 marks]

a) Describe how lanthanides are extracted from concentrated monazite mineral in acidic treatment process [15 marks]

b) Explain how Ce is obtained from the lanthanide [5 marks]

QUESTION THREE [20 marks]

a) List seven differences between lanthanides and actinides [14 marks]

b) Given that Gd^{+3} (Gd has configuration $[Xe]4f^7 5d^1 6s^2$) is colourless, Nd^{+3} (Nd has configuration $[Xe]4f^4 5d^0 6s^2$) is red, Predict the colours of the following ions:

(i) **U^{+3}**, U has a configuration of $[Rn]5f^3 6d^1 7s^2$) [2 marks]

(ii) **Cm^{+3}**, Cm has a configuration of $[Rn]5f^7 6d^1 7s^2$ [2 marks]

c) Arrange the following actinide ions in order of decreasing order:

An^{+3}, An^{+4}, AnO_2^{+} and AnO_2^{+2} [2 marks]

QUESTION FOUR [20 marks]

a) Determine the ground state term symbol for Sm^{3+} given that Sm has the configuration $[Xe]4f^6 5d^0 6s^2$ [8 marks]

b) Why do magnetic properties of Ln^{3+} ions depend entirely on ground states and not exited states? [3 marks]

c) Calculate the magnetic moment for a complex of Ho^{3+} such as $Ho(phen)_2(NO_3)_3$ given that ground state term symbol is $^5I_8$. [4 marks]

d) Explain why f – f transitions in the electronic spectra of lanthanide complexes are weaker than d – d transitions in the corresponding spectra of transition metal complexes. [5 marks]

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FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SCH 408: STATISTICAL THERMODYNAMICS

Date: 24th September, 2018 Time 12.00 - 3.00 pm

INSTRUCTIONS:

- Attempt ALL questions.

SCH. 408 – STATISTICAL THERMODYNAMICS

INSTRUCTIONS: Attempt all Questions

- Physical constants
- $L = 6.62x10^{-34}Js$
- $R=8.314\ Jk^{-1}mol^{-1}$
- $K= 1.381x10^{-23}Jk^{-1}$
- $N_A=6.022\ x\ 10^{23}mol^{-1}$

Question 1

a) A system consists of a set of uniformly spaced energy levels of separations $1x10^{-20}J$, starting from first level which has energy of 0.00 J. Eight particles are placed within this system with a total energy of $6\ x\ 10^{-20}$ J.

(i) How many possible states are there that satisfy these requirements? (8 marks)

(ii) Which state is the most probable (2 marks)

b) If there exist two exited states at energies of 0.72 and 1.24 kJ/mol above the ground state in a system, what would be the percentage of particles occupying each state at equilibrium when the temperature is 300 K? (4 marks)

Question 2

a) (i) Explain the significance of the partition function. (2 marks)

(ii) Derive the Boltzmann Distribution starting from

$$n_i = e^{\propto - \beta \varepsilon i}$$ (4 marks

(iii) Calculate the partition function at $T=35^0C$ for the system with the following set of energy levels.

E/(kJmol^{-1})	1.5	3.7	5.4	8.2	8.7	11.[illegible]
Degeneracy	5	1	5	3	7	3

(4 marks)

a) Show that for the molecule of mass, m, and free to move a container of volume, V_1, the translational partition function is given by

$$q_{Trans} = q_x.q_y.q_z$$

where x,y,z are the length of sides of the container. (4 marks)

Question 3

a) (i). Show that the internal energy change of a system is given by

$$u-u_o = \frac{RT^2}{q}\frac{dq}{dT}$$

where q is the partition function of a collection of particles. (6 marks)

b) (i). Calculate the change of internal energy associated with the three degrees of freedom for a mole of oxygen (O_2) molecules at 25° C. (5 marks)

(ii). Find the molar constant pressure heat capacity associated with three degrees of freedom. (3 marks)

Question 4

a) Starting with the equation

S= k lnW

Show that $S = \frac{U-U_o}{T} + Rlnq$ (6 marks) py of N_2 at 25°C and 1bar.

(2 marks)

lar entropy of N_2 at 25°C. (2 marks)

.6 x 10^{-47}kgm^2

rational contribution to the entropy of 1.0 mol of

. The vibrational energy spacing is 4.26 x 10^{-20}J

(2 marks)

Question 5

a) Show that for rotational motion and a linear molecule

$q_{rot} = \frac{T}{\theta_{rot}}$ where T is the Kelvin characteristic rotational temperature.

(4 marks)

b) Estimate the bond length of CO at 25° C given;

θ_{rot}=2.78 K (4 marks)

c) Deduce (i) the average rotational internal energy change per degree of freedom at 25° C. (3 marks)

(ii) Heat capacity, C_{V_1} for a mole and N_2 molecules. (2 marks)

(iii) What does the symmetry number ,δ, represent for a molecule. (1 mark)

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FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SCH 410: TECHNIQUES IN ORGANIC CHEMISTRY

Date: 17th September, 2018 Time 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other FOUR questions

REQUIREMENTS

- Calculator, Graph paper, IR correlation spectra chart, NMR chemical shifts table.

QUESTION ONE [14 MARKS]

a) Differentiate the following terms: **[6 marks]**

   i. Vapor pressure and boiling point:

   ii. Bathochromic shift and Hyperchromism:

   iii. Base peak and molecular ion peak:

b) Explain why simple distillation can be used to separate a mixture of methylene dichloride (bpt 40 ^{0}C and octane 125 ^{0}C while it cannot be used to separate a mixture of toluene (bpt 111 ^{0}C) and ethanol (Bpt. 78 ^{0}C. **[3 marks]**

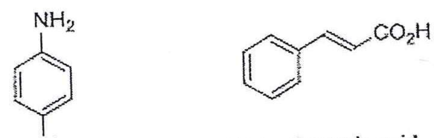
c) Arrange the following compounds in order of increasing Rf in a TLC analysis with silica gel as stationery phase: benzoic acid, benzaldehyde, nonane, and cyclohexanol. Explain the order. **[3 marks]**

d) The Rf of ibuprofen was found to be 0.32 when hexane was used as the development solvent. What effect would there be on the Rf of ibuprofen if acetone had been used to develop the TLC plate? **[2 marks]**

QUESTION TWO [14 MARKS]

a) An ideal binary solution of miscible liquids A and B has the following parameters: mole fraction of A (X_A) is 0.40 and the partial pressure of a pure liquid A (P_{Ao}) is 1710 mmHg, while the mole fraction of B (X_B) is 0.60 and the partial pressure of a pure liquid B (P_{Bo}) is 127 mmHg. Calculate the composition (in mole percentage) of the vapor from a distilling an ideal binary solution at 150 °C and 760 mmHg for the solution. **[4 marks]**

b) During an organic chemistry practical, a student was provided with a mixture of *p*-chloroanilline and *trans*-cinnamic acid in diethyl ether. Study the diagram and answer the questions that follow.



para-chloroaniline
mp 69-73 °C

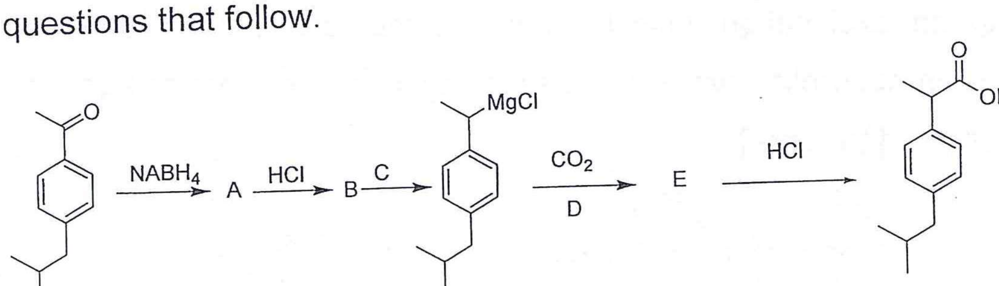
trans-cinnamic acid
mp 131-136 °C

i. Using equations explain stepwise how you would separate and recover the two compounds by extraction. **[8 marks]**

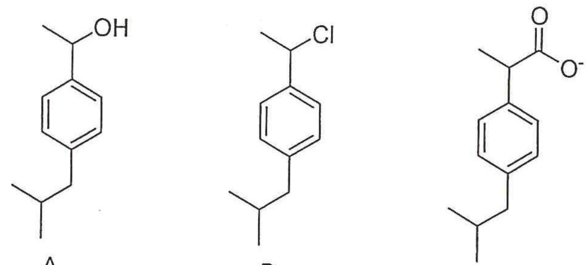
ii. After separation by extraction, explain how you would identify the each compound. **[2 marks]**

QUESTION THREE [14 MARKS]

In multistep synthetic reactions reactants are converted to intermediates which are then converted. The following scheme shows the pathway used by 4th year Organic chemistry student at Maseno University to the synthesis a pain killer referred to as ibuprofen. Study the scheme and answer the questions that follow.



a. Complete the scheme by drawing the structures of intermediates A, B and E. **[3 marks]**



A B E

b. Write the name of reagent C. **[1 mark]**

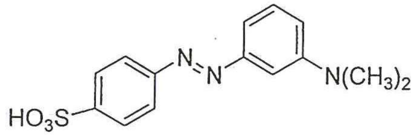
c. Suppose a student added water instead of carbon dioxide to the Grignard reagent in this experiment. Draw and name the products this student would obtain **[2 marks]**

d. During synthesis of ibuprofen, *p*-Isobutylacetophenone is used as one of the starting materials. Draw a scheme for converting benzene to *p*-Isobutylbutylacetophenone. Name the organic reagents and products at each stage. **[4 marks]**

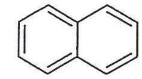
e. Sketch a gas chromatogram of a hypothetical solution that contains pentane, hexane and heptane dissolved in a suitable solvent. Show peaks for each compound on the chromatogram. **[4 marks]**

QUESTION FOUR [14 MARKS]

a) Giving an explanation indicate which of the following compounds shows an absorption within the visible region of the electromagnetic spectrum. **[4 marks]**

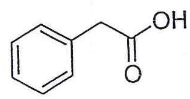


Compound 1


Compound 2

b) During an organic synthesis experiment synthesis compound 1, a student used the wrong reagents and obtained compound 2. By use of IR correlation tables explain how IR spectroscopy could be used to distinguish between the two compounds?

[6 marks]



Compound 1

Compound 2

c) Calculate the concentration and transmittance of a solution if the molar absorption coefficient of the compound at 280 nm is 75 $mol^{-1}dm^{3}cm^{-1}$, a 1 cm cuvette is used and an absorbance of 0.45 is obtained during analysis. **[4 marks]**

QUESTION FIVE [14 MARKS]

a) Define the following phrases as used in IR spectroscopy. **[4 marks]**

i. A stretching vibration:

ii. A deformation:

iii. Translation motion:

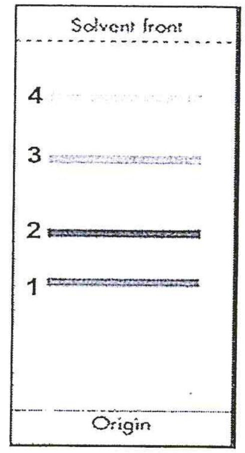
iv. Rotation of the molecule:

b) Briefly describe the steps a sample undergoes during mass spectrometry analysis. **[4 marks]**

c) Outline the steps followed when using GC to monitor the course of an organic reaction. **[6 marks]**

QUESTION SIX (14 MARKS)

a) An extract from a plant was analysed using thin-layer chromatography with a non-polar solvent. The chromatogram obtained is shown in the figure below.



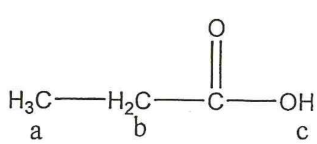
The table gives the R_f values of some materials commonly found in plants.

Chemical	R_f
Xanthophyll	0.67
β-carotene	0.82
Chlorophyll *a*	0.48
Chlorophyll *b*	0.35
Leutin	0.39
Neoxanthin	0.27

a) Identify the different components by completing the following table.

[4 marks]

b) Draw a sketch showing the proton NMR splitting patterns (singlet, doublet, etc.) for each methylene group in the structure below. Explain how you determined each splitting pattern. **[6 marks]**



c) Briefly describe how the following detectors function in gas chromatography. **[6 marks]**

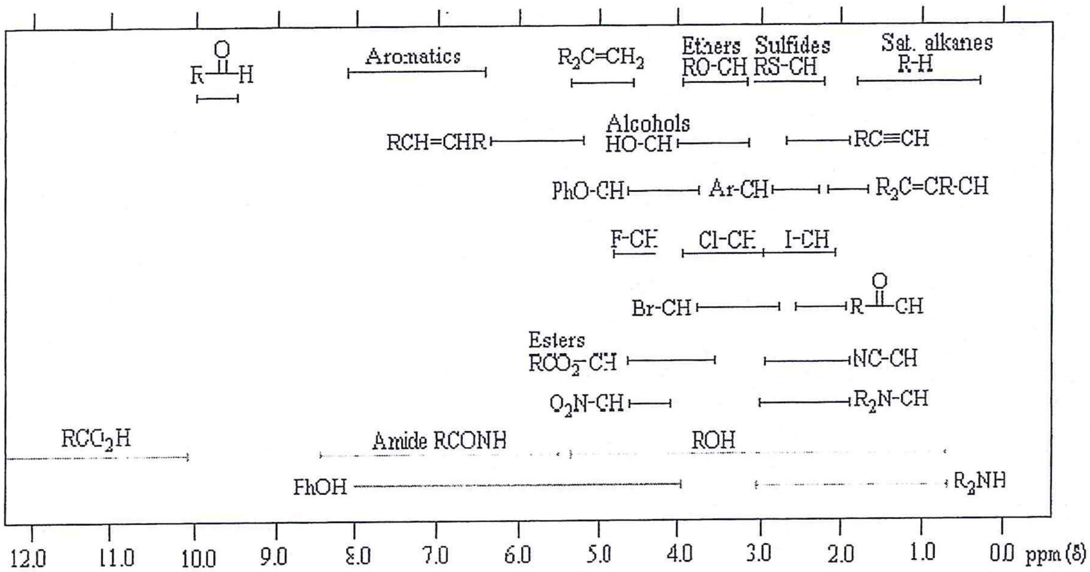
i. Flame ionization detectors:

ii. Thermal conductivity detectors:

IR: Correlation table for characteristic Group Frequencies of Organic Molecules

Frequency, cm^{-1}	Bond	Functional group
3640–3610	O–H, free hydroxyl	alcohols, phenols
3500–3200	O–H, H–bonded	alcohols, phenols
3400–3250	N–H	primary, secondary amines, amides
3300–2500	O–H	carboxylic acids
3330–3270	—C≡C—H: C–H	alkynes (terminal)
3100–3000	C–H	Aromatics
3100–3000	=C–H	Alkenes
3000–2850	C–H	Alkanes
2830–2695	H–C=O: C–H	Aldehydes
2260–2210	—C≡N	Nitriles
2260–2100	—C≡C—	Alkynes
1760–1665	C=O	carbonyls (general)
1760–1690	C=O	carboxylic acids
1750–1735	C=O	esters, saturated aliphatic
1740–1720	C=O	aldehydes, saturated aliphatic
1730–1715	C=O	alpha, beta–unsaturated esters
1715	C=O	ketones, saturated aliphatic
1710–1665	C=O	alpha, beta–unsaturated aldehydes, ketones
1680–1640	–C=C–	Alkenes

1650–1580	N–H	primary amines
1600–1585	C–C (in–ring)	Aromatics
1550–1475	N–O	nitro compounds
1500–1400	C–C (in–ring)	Aromatics
1470–1450	C–H	Alkanes
1370–1350	C–H	Alkanes
1360–1290	N–O	nitro compounds
1335–1250	C–N	aromatic amines
1320–1000	C–O	alcohols, carboxylic acids, esters, ethers
1300–1150	C–H (–CH_2X)	alkyl halides
1250–1020	C–N	aliphatic amines
1000–650	=C–H	Alkenes
950–910	O–H	carboxylic acids
910–665	N–H	primary, secondary amines
900–675	C–H	Aromatics
850–550	C–Cl	alkyl halides
725–720	C–H	Alkanes
700–610	$-C\equiv C-H$: C–H	Alkynes
690–515	C–Br	alkyl halides



^{1}H NMR Chemical shifts

MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SCH 411: ORGANIC STEREOCHEMISTRY

Date: 27th September, 2018 Time 3.30 - 6.30 pm

INSTRUCTIONS:

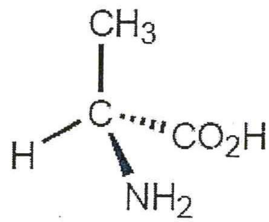
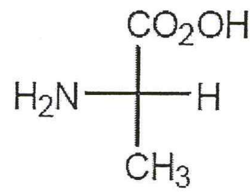
- Answer ANY FIVE questions.

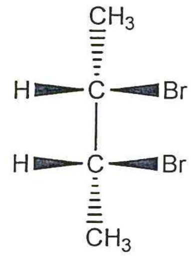
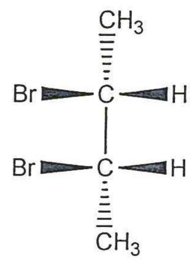
SCH 411: Organic Stereochemistry Examination for 2017/2018

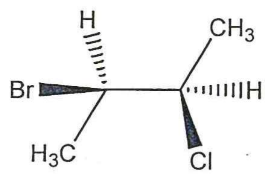
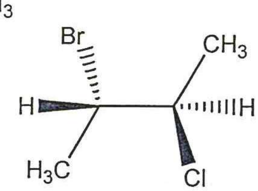
Instruction
Attempt any five (5) questions

Question one

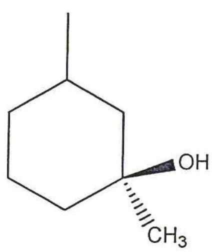
A. Which of the following terms best describes the pair of compounds shown below: enantiomers, diastereomers, or the same compound? **[3 marks]**

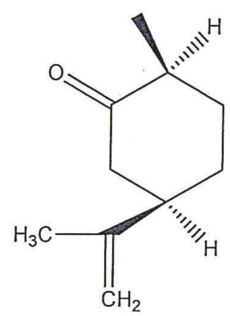
(i)  

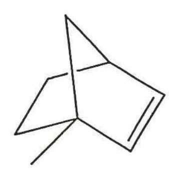
(ii)  and 

(iii)  and 

B. Redraw the compounds and label each asymmetric carbon in the compound below as *R* or *S*. **[5 marks]**

(i) 

(ii) 

(iii) 

C. Draw the structure of the following compounds and TAKE PARTICULAR NOTE to indicate three dimensional stereochemical details properly. **[3 marks]**

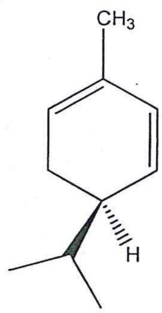
(i) (2*R*,3*S*)-2,3-dichloropentane
(ii) (*S*)-1-bromo-1-chloropropane
(iii) a *meso* form of 1,3-dichlorocyclopentane.

SCH 411: Organic Stereochemistry Examination for 2017/2018

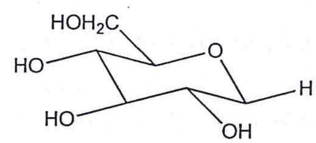
Identify the number of asymmetric carbons present in the compounds below.

[3 marks]

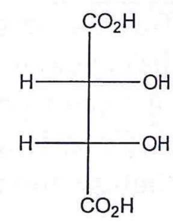
(i)



(ii)

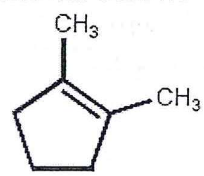


(iii)



Question two

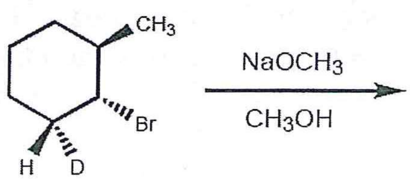
A. One of the products that results when 1-bromo-2,2-dimethylcyclopentane is heated in ethanol is shown below.



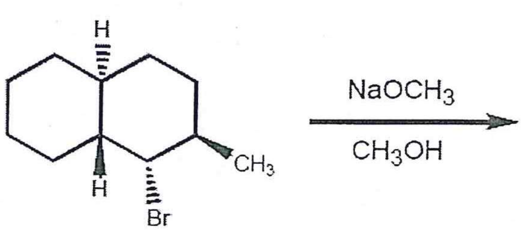
(i) Give the name of the mechanism leading to that product. **[2 marks]**

(ii) Draw that mechanism involved, TAKE PARTICULAR NOTE to indicate stereochemistry properly. **[3 marks]**

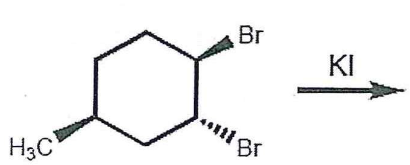
B. Provide the structure of the major organic product in the following reaction schemes.



(i) **[1 mark]**



(ii) **[1 mark]**

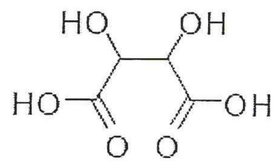


(iii) **[1 mark]**

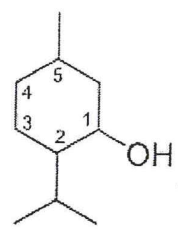
C. Which diastereomer of 1-bromo-4-*t*-butylcyclohexane, the *cis* or the *trans*, undergoes elimination more rapidly when treated with sodium ethoxide ($CH_3CH_2O^-Na^+$)? Explain your answer. **[6 marks]**

Question three

A. The Pasteur separation of the mirror-image crystals of tartaric acid was one of the developments that led to the understanding of chirality in molecules. Draw all stereoisomers for tartaric acid and fully indicate the interrelationships of all (enantiomers, diastereomers or meso compounds). Be sure to assign *R*, *S* configurations to each chiral center. **[7 marks]**



B. Menthol is a member of the terpene family of natural products. It exists naturally in a (*1R,2S,5R*) form and a (*1S,2R,5S*) forms (ring numbering is shown).



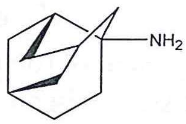
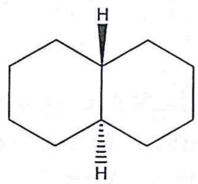
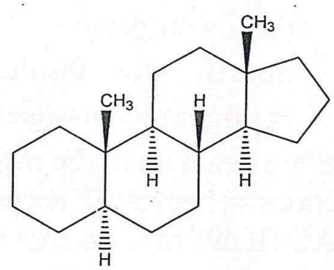
(i) Are these two compounds *enantiomers* or *diastereomers*? **[1 mark]**

(ii) Draw the two molecules in their chair conformations and identify their most stable chair conformer(s). **[6 marks]**

SCH 411: Organic Stereochemistry Examination for 2017/2018

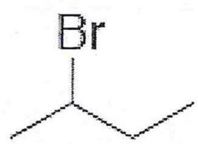
Question four

A. Draw a three-dimensional representation of following compounds showing the chair cyclohexane rings. **[3 marks]**

(i)  (ii)  (iii) 

B. *Cis*-1,2-Dimethylcyclopropane has more strain than *trans*-1,2-dimethylcyclopropane. Draw the two isomers and account for this difference in strain energy. **[4 marks]**

C. Consider 2-bromobutane (shown below). Sighting along the C2-C3 bond and using Newman projections:



Interactions	$Kcal^{-1}mol^{-1}$
H,H eclipsed	1.0
H-CH_3 eclipsed	1.4
CH_3-CH_3 eclipsed	2.6
H-Br eclipsed	1.7
CH_3-Br gauche	0.9
CH_3-CH_3 gauche	0.9
CH_3-Br eclipsed	3.8

(i) Draw the most stable conformer and calculate the total interaction energy. **[3 marks]**

(ii) Draw the least stable conformer and calculate the total interaction energy. **[3 marks]**

(iii) Determine the total barrier to rotation between the two conformers. **[1 marks]**

Question five

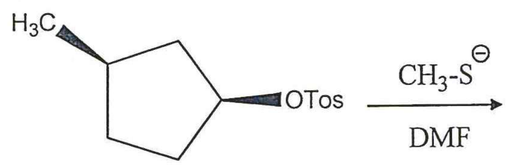
A. When (R)-1-bromo-2-butanol reacts with KI in acetone the product is 1-iodo-2-butanol.

(i) Draw the structure of 1-iodo-2-butanol indicating the stereochemical details properly. **[1 mark]**

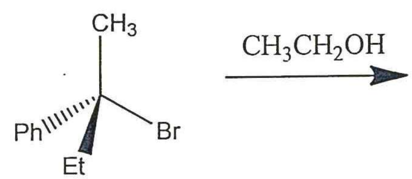
(ii) Specify the absolute configuration of the product. Justify the configuration assigned. **[2 marks]**

B. Give the structure of the major product(s) expected from each of the following reactions schemes. If necessary, indicate the product stereochemistry. "NO REACTION" may be an acceptable answer. **[8 marks]**

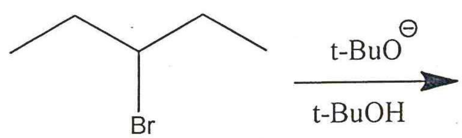
(i)



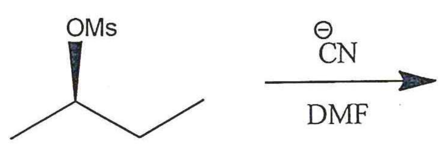
(ii)



(iii)



(iv)



Questions six

A. A sample of 2-methyl-1-butanol has a specific rotation, $[\alpha]_D^{25}$, equal to +1.151°.

(i) What is the % enantiomeric excess of the sample given that a pure sample of the (S) isomer has a specific rotation of –5.756°? **[2 marks]**

(ii) Which enantiomer is in excess, the (R) or the (S)? **[2 marks]**

(iii) What is the percentage composition of each form? **[3 marks]**

B. You have a sample (Sample X) which is a mixture of +/- Carvone. The solution was made by dissolving 4.50 g of the sample in enough methanol to bring the volume of solution to 10.0 mL. Some of the solution is placed in a 100 cm polarimeter cell and its optical rotation is measured at 25°C using light of the sodium D line wavelength (589.6 nm). The observed rotation is +22.2°.

(i) What is the specific rotation of this sample? **[2 marks]**

(ii) An enantiomerically pure sample of the S enantiomer of Carvone has a specific rotation of +15.5°. What is the % enantiomeric excess of Sample X? **[2 marks]**

(iii) What is the composition of Sample X? **[3 marks]**

MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SCH 412: QUANTUM CHEMISTRY II

Date: 21st September, 2018 Time 8.30 - 11.30 am

INSTRUCTIONS:

- Attempt ANY FIVE questions.

SCH 412: QUANTUM CHEMISTRY II

<u>**INSTRUCTIONS TO CANDIDATES**</u>**: Attempt ANY FIVE questions**

1. The state of a quantum mechanical system is completely specified by a function $\psi(r,t)$ that depends on the coordinates of the particle(s) and on time.
   a) What is the important physical meaning of the wave function? [2 marks]
   b) Which mathematical conditions are imposed on the wave function? [4 marks]
   c) Define the following concepts: [2 marks]
      (i) Operator
      (ii) Eigenvalue
   d) How is an operator related to its eigenvalue? [3 marks]
   e) Show how quantum mechanics determines the outcome of an experiment, e.g., frequency of radiation. [3 marks]

2. The normalized 1s wave function for a one-electon atom in atomic units is

$$\Psi_{100}(r,\theta,\phi)=\frac{Z^{3/2}}{\sqrt{\pi}}e^{-Zr}$$

where Z is the atomic number.
   a) Write the general expression for the operator that would allow you to calculate the average value of a quantum mechanical property. [2 marks]
   b) The expectation value of what operator would correspond to the average value, in a.u., that you would obtain after a very, very large number of measurements of the distance of the 1s electron from the nucleus in the hydrogen atom? [3 marks]
   c) Would the value for the same measurement be larger or smaller for He^{+}? Explain your answer. [2 marks]

   d) Prove that the average distance of the 1s electron from the proton in the hydrogen atom is 1.5 a.u. You may find the following integral useful

$$\int_0^\infty r^2 e^{-2r} dr = \frac{n!}{2^{n+1}}$$

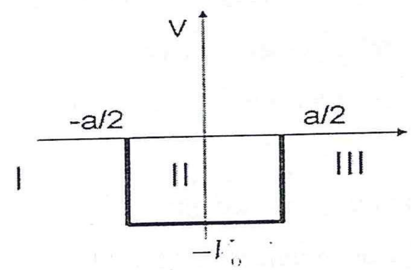
You may also find it useful to recall that the spherical polar volume element is $r^2 dr \sin\theta d\theta d\varphi$. [2 marks]

e) A particle in a one-dimensional box of length L has an energy

$$E = \frac{2\hbar^2\pi^2}{mL^2}$$

What is the probability that a measurement of the location of the particle will find it between $x = 0$ and $x = L/2$? [5 marks]

3. Figure 1 is a finite potential well of depth V_o



Write the schrodinger equation and its appropriate wave function for the three space regions marked I, II and III. Evaluate the total energy in each case. [14 marks]

4. a) What is the value of the integral, $\int_0^L \psi_m(x)\,\psi_n(x)dx$ for the allowed wavefunctions of a particle in a one dimentional box (PIAB)? [2 marks]

b) The Schrödinger equation for the PIAB can be compactly expressed as

$$\hat{H}\psi_n(x) = E_n\psi_n(x)$$

where $\psi_n(x)$ and E_n are the allowed wavefunctions and energies.

(i) What is meant when we refer to the E_n as the "allowed" energies? [2 marks]

(ii) What do we mean by the "allowed" wave functions, $\psi_n(x)$? [2 marks]

(iii) Comment on the possibility of determining a specific value of energy, say E_2 by experiment? [2 marks]

c) A hydrogen atom is in the 3D state ($n = 3, l = 2$)

(i) What are the possible values of j? [2 marks]

(ii) What are the possible magnitude of the total angular momentum? [2 marks]

(iii) What are the possible z components of the total angulsr momentum? [2 marks]

5. a) What is the Born-Oppenheimer approximation? [4 marks]

b) Write the electronic Schrödinger equation. [2 marks]

c) The excited electronic configuration of helium atom can exist either as a singlet and a triplet state.

(i) Tell which state has the lower energy and explain why. [3 marks]

(ii) Give an expression which represents the energy separation between the singlet and triplet states in terms of the one –electron orbitals ψ_{1s} and ψ_{2s}. [5 marks]

6. a) Formulate the variational principle / theorem. [2 marks]

b) The roadmap to solving the Schrödinger equation for molecules consist of three parts: Briefly explain what is done in each part. [6 marks]

c) If a H_2 molecule rotates in the plane of a crystalline surface (in a chemisorption situation), it can be approximated as a two-dimensional rigid rotor. Calculate (in kJ/mol) the lowest energy rotational transition for such a system. Take the mass of the rotor to be 1/2 the mass of a hydrogen atom with a value of r equal to the bond length of H_2, 0.7416 Å. Clearly show your working with appropriate conversion units. [6 marks]

... end...

MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FOURTH YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION TECHNOLOGY

MAIN CAMPUS

SCH 415: POLYMER CHEMISTRY

Date: 21st September, 2018 Time 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer ANY FIVE questions.

SCH 415: POLYMER CHEMISTRY

Answer any five questions

1 (a) Trace amounts of hydrogen peroxide initiates polymerisation of acrylonitrile (CH_2=CHCN) to produce Orlon.

i. Write a detailed mechanism of the polymerisation process, explaining each step of the reaction **(5 marks)**

ii. Discuss the fate of the reaction if the solvent was contaminated with tetrabromomethane. **(2 marks)**

(b) In the presence of naphthalene, sodoum catalyses the polymerisation of vinyl chloride (CH_2=CHCl)

i. Write detailed mechanism of the polymerisation process, explaining each step. **(5 marks)**

ii. Suggest how this polymerisation process can be terminated. **(2 marks)**

2 (a) i. Draw the structures of gutta percha and natural rubber. **(2 marks)**

ii. Between the two polymers, which one is elastic? **(2 marks)**

iii. Explain why it is elastic while the other is non-elastic **(3 marks)**

iv. Give one industrial use of gutta-percha **(2 marks)**

(b) The elastic rubber in 2 (a) i. can be strengthened by cross-linking,

i. Explain how this is done industrially. **(3 marks)**

ii. Draw the structure of the strengthened rubber **(2 marks)**

3 Ethyl titanium trichloride catalyses the polymerisation of cis and trans 2-butene.

i. What is the name of this polymerisation process? **(2 marks)**

ii. Propose a mechanism of the polymerisation process using one of the two butanes. **(5 marks)**

iii. Draw three dimensional structures of the polymer formed from each butane. **(2 marks)**

iv. Explain the difference(s) if any, between the polymers produced, and that which could have been produced if the butenes were polymerised in the presence of hydrogen peroxide. **(5 marks)**

4(a) Cyclopentadiene reacts with maleic anhydride to form compound B. When treated with 1,6-haxanediamine in the presence of sodium acetate B produces a polymer.

i. Give the structure of compound B. **(2 marks)**

ii. Give the structure of the polymer formed. **(3 marks)**

iii. Discuss what would happen if the solvent used in the polymerisation reaction was contaminated with hydrogen peroxide. **(3 marks)**

(b) Cellulose reacts with *betta*-propionate (propyl lactone) or *gamma*-butyl lactone to form graft polymers.

i. Write structures of the polymers formed. **(4 marks)**

ii. Explain the differences in the polymerisation reactions if any. **(2 marks)**

5(a) Using appropriate illustrations, explain the difference between these terminologies:

i. Addition polymerisation and condensation polymerisation **(2 marks)**

ii. Graft polymer and block polymer **(2 marks)**

iii. Homopolymer and copolymer **(2 marks)**

iv. Syndiotactic polymer and atactic polymer **(2 marks)**

v. Thermoplastic polymer and thermosetting polymer **(2 marks)**

(b) Discuss a possible use of hydroxyl propyl cellulose in industrial waste management. **(4 marks)**

6 (a) 2-Methly phenol and 3-methyl phenol each reacts with formaldehyde in the presence hydrochloric acid to produce polymers.

i. Draw the structures of the polymers produced from the two phenols. **(3 marks)**

ii. Propose a reasonable mechanism of the polymer formation using 4-methlyphenol as your monomer. **(4 marks)**

(b) Phenyl isocyanate reacts with water to produce unstable compound **X**. The formed compound breaks down to produce carbon dioxide and a compound which reacts with phenyl isocyanate to produce **Y**. **Y** reacts with formaldehyde to produce a polymer.

i. Give the structure of compound **X**. **(2 marks)**

ii. Name compound **Y**. **(2 marks)**

iii. Write the structure of the polymer produced. **2 marks)**